

How Chatbots are Enabling a Paradigm Shift for Organisations

When a human resource is unable to handle the continuous demand for customer service well, organisations switch to AI-powered chatbots. These chatbots are now being used in various fields to improve customer service and thus the revenue.

“We are now witnessing a new shift in computing: the move from a mobile-first to an AI-first world.”

—Sundar Pichai, CEO of Google

The advances in technology have increased exponentially, which has brought a revolution in the user interface by extracting maximum value from artificial intelligence (AI) to enhance the productivity of businesses and boost their revenue. In recent years, the intense innovations in machine learning and other fundamental technologies like natural language processing have led to an astonishingly widespread use of chatbots or conversational agents that facilitate the human-to-bot interaction, which is the biggest dominating mobile trend of this time.

Chatbots, derived from ‘chat robots,’ are text- or voice-based algorithms powered by AI that are capable of checking the customers’ needs and providing possible suggestions. These are now gaining more popularity on Facebook Messenger, Twitter, LinkedIn, Skype, Slack, Kik, and other messaging platforms.

The human-to-bot interaction lets the users have normal conversation in dialogue form, entering their purposes directly into the machines to buy things, to book their place in the salon, to make hotel reservation, to get online health advice, or to even get directions. The chatbot appears as if the machine

language has been humanised and is making the computers or mobile phones adopt the human world. But, in fact, the human language is being mechanised to bring us into the machine world!

Chatbots, virtual assistants and live chat software

Chatbots and virtual assistants (VA) are both smart applications of AI as well as conversational interfaces. Chatbots are automated software that is company-oriented, programmed with a learned set of knowledge and reacts to users’ questions by providing a pre-defined answer. Whereas VAs like Cortana or Siri are digital assistants, similar to human assistants, that deal with an individual user, like organising and reminding business appointments, handling to-do lists, setting up alarms, and assisting in other daily routines.

Chatbots, being information acquisition tools, facilitate customers’ communication with any business venture. An organisation can build its customised chatbot for virtual conversation with its customers using a technology called Natural Language Processing (NLP). The NLP algorithm is making the chatbot smarter.

VA uses high-level automation software, based on advanced NLP, which enables it to understand natural language voice commands and interact with the individual user, offering human-like assistance in a highly effective way. VA follows the dynamic conversational flow



Figure 1: Chatbots in e-commerce business